Matthew Penner

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EDUCATION

Portland State University Portland, OR
Bachelor of Science in Computer Science (University Honors College) Sept. 2021 – Dec. 2025
Thesis: Experience from a Distance - Improving Transparency for the Multnomah Athletic Club

- Collaborated with a six-person team to deliver a Senior Capstone project addressing remote marketing challenges
- Developed a web application featuring 360-degree photography and a dynamic interactive mapping system

EXPERIENCE

SII Group USA Inc. **Austin, TX**
QA Automation Engineer *July 2026 – Present*

- Engineered test frameworks: Built robust automated testing architectures using Python, pytest, and Playwright.
- Accelerated release cycles: Developed end-to-end UI and API test suites that significantly reduced manual testing overhead.
- Integrated QA processes: Collaborated across engineering teams to embed quality standards directly into the development lifecycle.
- Managed defects: Identified. tracked. and resolved complex software bugs in close coordination with cross-functional teams.
- Optimized efficiency: Refined existing test scripts and automation architecture to improve overall testing performance.

Intel **Hillsboro, OR**
Inventory Technician Intern *May 2023 – July 2026*

- Created custom scripts to streamline work processes and conducted audits of inventory rooms for Ethernet board parts
- Responsible for worldwide shipping and transporting electronic components to manufacturing warehouses
- Received a letter of recognition from management for quality control and reporting excellence

PROJECTS

Lunar Gateway Autonomy
Go, Python, FastAPI, React, Kubernetes, MQTT

- Developed a high-reliability edge controller and autonomous AI decision engine to manage simulated space station operations
- Built a Go-based telemetry ingestion module utilizing MQTT for power management and thermal control load shedding
- Designed a React and TypeScript visualizer to track AI choices, monitor the power grid, and parse structured cargo logistics

UAV Flight Optimizer
Go, Python, React, TypeScript, MapboxGL

- Engineered a physics and pathfinding engine utilizing a custom 3D A* algorithm weighted by drone energy consumption
- Created a Go environment service to fetch real-time 3D wind vectors and parse GeoTIFF topographic elevation data
- Developed an interactive mission planner frontend to visualize 3D flight paths, altitude profiles, and estimated battery drain

TECHNICAL SKILLS

Languages: C, C++, Python, Java, TypeScript, JavaScript, SQL, Rust, Go, Kotlin
Frameworks/Technologies: React, Node.js, Express, React Native, Prisma ORM, Docker, Redis
Systems/Tools: Linux, GDB, Git, Terraform, AWS, GitHub Actions, PostgreSQL, MongoDB
Concepts: Multithreading (POSIX), Concurrency, TCP/IP Networking, Unit Testing, RESTful APIs

ACTIVITIES & LEADERSHIP

Civil Air Patrol **Hillsboro, OR**
Cadet Technical Sergeant *2016 – 2020*

- Participated in the Air Force's annual National CyberSecurity competition
- Volunteered for community site security and completed orientation flight lessons

Computer Action Team **Portland, OR**
Front Desk Volunteer *2023 – 2024*

- Responded to tickets and proxy tickets
- Assisted other students and faculty in software issues and administrative tasks